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Team Members:

Tirta Aji

Farid Al Ayubi

Jason Frederick K

Daffa F.

Contact email: tirtaajin33@gmail.com

Project information

- Goal

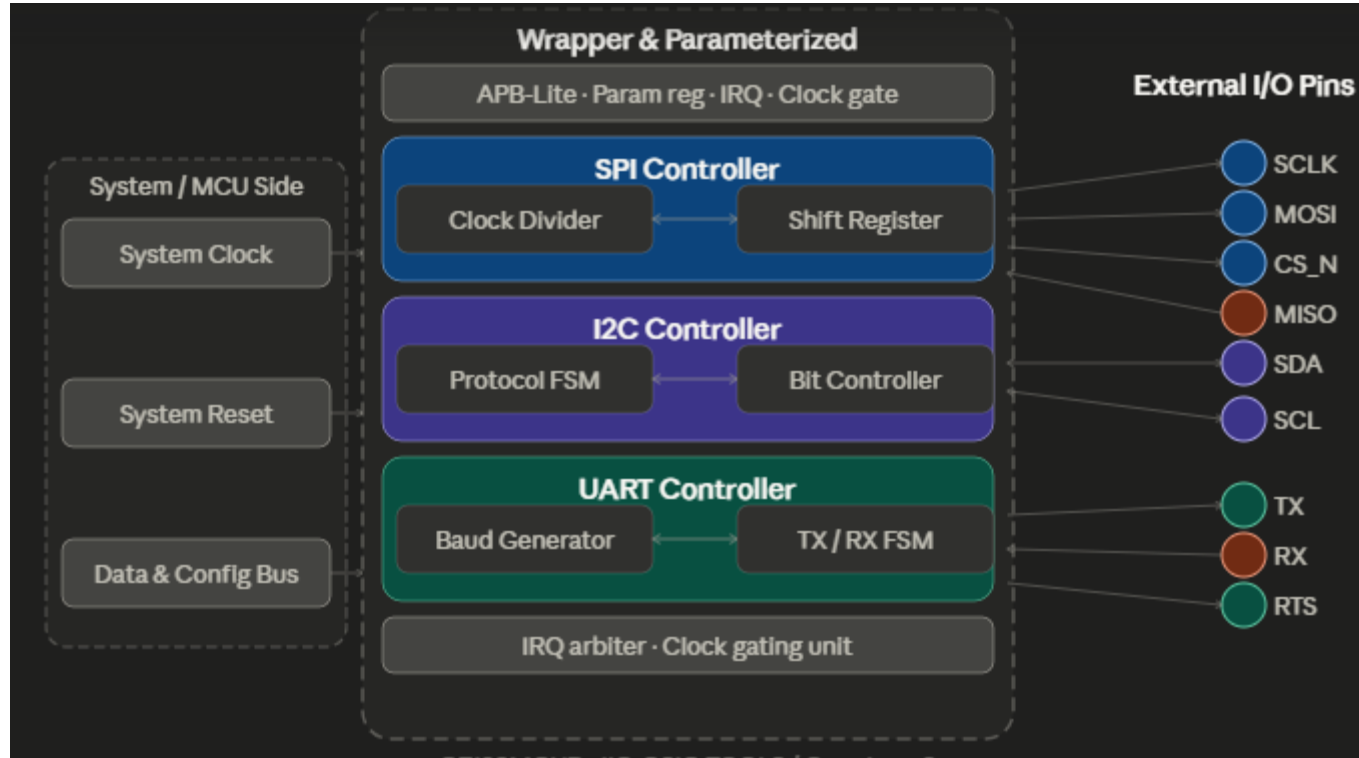
This project aims to design, verify, and tape out a fully open-source, parameterized serial communication IP block — integrating UART, I2C, and SPI controllers — targeting the GlobalFoundries GF180MCUD process node using the IIC-OSIC-TOOLS and OpenLane2 digital design flow. By exposing a rich set of compile-time Verilog parameters (such as protocol enable flags, FIFO depth, data-path width, and addressing mode) alongside a runtime-configurable APB-Lite register interface, the IP is designed to serve as a reusable, community-grade building block that any open-source SoC integrator can instantiate and customize without modification to the core RTL.

Project Overview

- Design – High Level Proposal

The IP is structured into four independent Verilog modules: a UART block (8N1/8N2, programmable baud divider, parameterized TX/RX FIFO depth, optional RTS/CTS flow control), an I2C block (7-/10-bit addressing, 100 kHz / 400 kHz / 1 MHz SCL, optional slave mode and multi-master arbitration), an SPI block (full-duplex, modes 0–3, configurable word width up to 32 bits, up to four chip-select lines), and a top-level Wrapper & Parameterized block that instantiates all three through Verilog generate constructs, exposes a single APB-Lite slave port to the host processor, arbitrates interrupts, and provides per-protocol clock gating. Compile-time parameters govern structural decisions that become permanent at tapeout, while runtime parameters such as baud-rate divisor, SPI clock polarity/phase, I2C SCL divider, and interrupt watermarks are stored in a memory-mapped register file and can be written by firmware at any time after power-on.

Project Information



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- Application

This controller acts as a reliable digital bridge for embedded systems, enabling seamless data acquisition from various external peripheral modules such as industrial field sensors and debugging terminals.

Team background

Our team is composed of three third year and one second year undergraduate Electrical Engineering Students from Diponegoro University, Indonesia. **We have no prior IC design Experience.**

Questions, Suggestions, Doubts?